Begonnen am	Montag, 22. Januar 2024, 12:02
Status	Beendet
Beendet am	Montag, 22. Januar 2024, 12:12
Verbrauchte Zeit	10 Minuten
Bewertung	**10,00** von 10,00 (**100**%)

Frage 1
Richtig

Erreichte Punkte 1,00 von 1,00

How to implement the **inner product** (`c`) of `a` and `b` represented by the for-loop below, using `numpy`?

```
a, b = [1, 2, 3], [4, 5, 6]
c = sum(ai * bi for ai, bi in zip(a, b))
```

Note: there are multiple correct answers, but you only need to pick one!

- ○ `c = np.prod(a, b)`
- ○ `c = np.array(a) * np.array(b)`
- ○ `c = np.inner_product(a, b)`
- ○ `c = np.matmul(a, b)`
- ◉ `c = np.array(a) @ np.array(b)` ✔

Your answer is correct.

Frage 2
Richtig

Erreichte Punkte 2,00 von 2,00

Which operation on matrices `a` (with shape `(K, M)`) and `b` (with shape `(K, N)`) would produce an output with shape `(N, M)`?

Note: there are multiple correct answers, but you only need to pick one!

Wählen Sie eine Antwort:

- ○ `a @ b.T`
- ○ `np.dot(a, b)`
- ○ `np.matmul(a.T, b).T`
- ○ `np.matmul(a, b, shape=(N, M))`
- ◉ `np.transpose(b) @ a` ✔

Your answer is correct.

Frage **3**

Richtig

Erreichte Punkte 1,00 von 1,00

What loss function would you use if the goal is to maximise the likelihood of a dataset with target values distributed according to a Gaussian distribution?

In other words, which loss function do you need to minimise in order to maximise $\mathcal{L}(\theta\,;\boldsymbol{x},\boldsymbol{y}) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(g(\boldsymbol{x}\theta)-\boldsymbol{y})^2}{2\sigma^2}\right)$?

Wählen Sie eine Antwort:

a.
```
def average_error(predictions, targets):
    return np.mean(predictions * targets, axis=-1)
```

b.
```
def cross_entropy(predictions, targets):
    return -np.sum(targets * np.log(predictions), axis=-1)
```

c. (selected) ✔
```
def squared_error(predictions, targets):
    return (predictions - targets) ** 2 / 2.
```

d.
```
def logit_error(predictions, targets):
    return np.sum(np.log(predictions - np.max(predictions)) - targets, axis=-1)
    return
```

e.
```
def logistic_error(predictions, targets):
    return -targets * np.log(predictions) - (1 - target) * np.log(1 - predictions)
```

Your answer is correct.

Frage **4**

Richtig

Erreichte Punkte 1,00 von 1,00

What is numerical differentiation typically used for when developing a library like `nnumpy`?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Finding errors in the forward pass of the network.
- [x] b. Testing the gradient computations during development. ✔
- [ ] c. Computing the gradients for updating the network parameters.
- [ ] d. Checking the statistics of the gradient computations.
- [ ] e. Numerical differentiation has no function when implementing neural networks

Your answer is correct.

Frage 5
Richtig
Erreichte Punkte 1,00 von 1,00

What `nnumpy` module implements the functionality of function composition, i.e. $\boldsymbol{f} : \mathbb{R}^N \rightarrow \mathbb{R}^M : \boldsymbol{x} \mapsto \boldsymbol{y} = \mathrm{layer}_2(\mathrm{layer}_1(\boldsymbol{x}))$?

Wählen Sie eine Antwort:

- a. `Sequential` ✔
- b. `ModuleList`
- c. `Linear`
- d. `MultiLayer`
- e. `MLP`

Your answer is correct.

Frage 6
Richtig
Erreichte Punkte 1,00 von 1,00

Given a 1D input signal, $\boldsymbol{x} = (4, -5, 1)$, and the 1D kernel $\boldsymbol{k} = (-5, -2, 1)$, what is the result of computing the cross-correlation $\boldsymbol{k} \star \boldsymbol{x}$?

- a. (1, 10, -20)
- b. (-20, 10, 1)
- c. (4, 10, -5)
- d. -20 + 10 + 1 ✔
- e. 4 + 10 + -5

Your answer is correct.

Frage 7
Richtig
Erreichte Punkte 2,00 von 2,00

How would you describe the computations of the gradients of a convolutional layer w.r.t. kernel weights, if the forward pass is implemented by means of convolution (**not** cross-correlation).

Wählen Sie eine Antwort:

- a. The convolution between gradients and kernel weights
- b. The cross-correlation between gradients and kernel weights
- c. The convolution between inputs and gradients ✔
- d. The cross-correlation between inputs and gradients
- e. The sum of gradients

Your answer is correct.

Frage 8

Richtig

Erreichte Punkte 1,00 von 1,00

How large is the `Adamax` optimiser state when training a network with a total of 98 parameters **without** bias correction?

- a. 98 values
- b. 9604 values
- c. 0 values
- d. 197 values
- e. 196 values ✔

Your answer is correct.